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COMPUTER NETWORKS

TEAM NETWORKS

Department of Computer Science and Engineering

Unit – 4 Network Layer and Internet Protocol

4.1 Overview of Network Layer

4.2 What's Inside a Router?

4.3 Switching

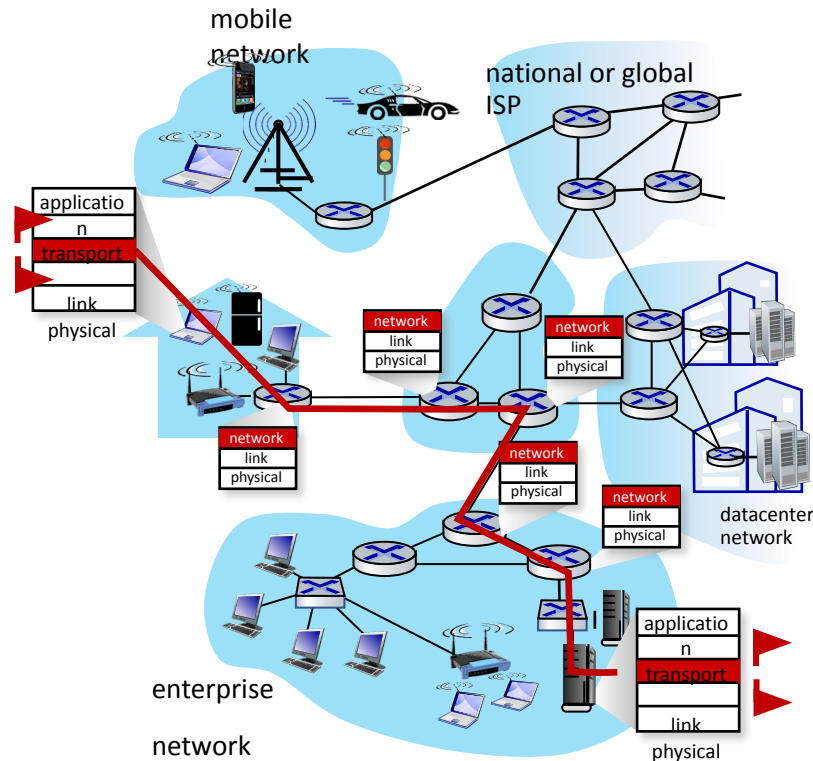
4.4 The Internet Protocol (IP)

- Datagram format
- Fragmentation
- IPv4 addressing
- NAT

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Network-layer Services and Protocol

- transport segment from sending to receiving host
 - **sender:** encapsulates segments into datagrams, passes to link layer
 - **receiver:** delivers segments to transport layer protocol
- network layer protocols in *every Internet device*: hosts, routers
- **routers:**
 - examines header fields in all IP datagrams passing through it
 - moves datagrams from input ports to output ports to transfer datagrams along end-end path



network-layer functions:

- *forwarding*: move packets from a router's input link to appropriate router output link
- *routing*: determine route taken by packets from source to destination
 - *routing algorithms*

analogy: taking a trip

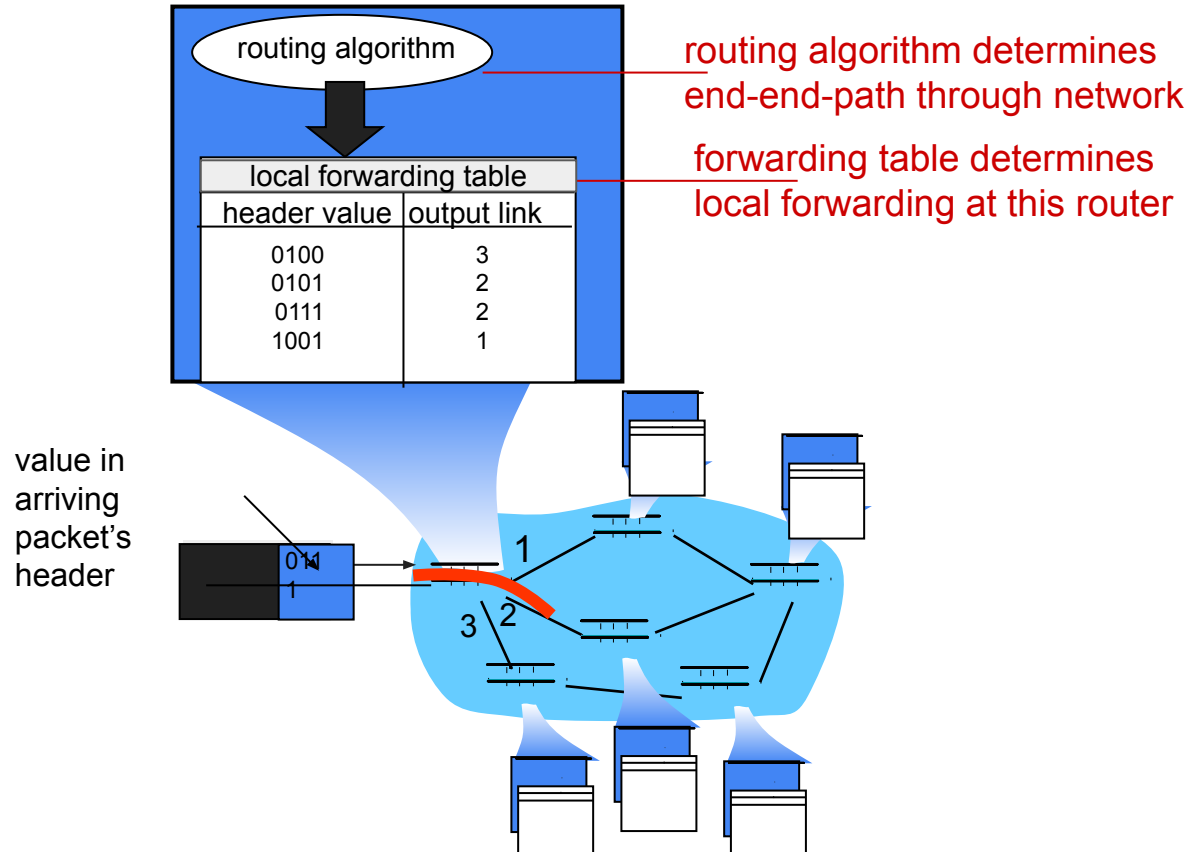
- *forwarding*: process of getting through single interchange
- *routing*: process of planning trip from source to destination



forwarding



routing



- 3rd important function in *some* network architectures:
 - ATM, frame relay, X.25
- before datagrams flow, two end hosts *and* intervening routers establish virtual connection
 - routers get involved
- network vs transport layer connection service:
 - *network*: between two hosts (may also involve intervening routers in case of VCs)
 - *transport*: between two processes

Q: What *service model* for “channel” transporting datagrams from sender to receiver?

example services for *individual* datagrams:

- guaranteed delivery
- guaranteed delivery with less than 40 msec delay

example services for a *flow* of datagrams:

- in-order datagram delivery
- guaranteed minimum bandwidth to flow
- restrictions on changes in inter-packet spacing

Network Architecture	Service Model	Quality of Service (QoS) Guarantees?			
		Bandwidth	Loss	Order	Timing
Internet	best effort	none	no	no	no

Internet “best effort” service model

No guarantees on:

- successful datagram delivery to destination
- timing or order of delivery
- bandwidth available to end-end flow

Network Architecture	Service Model	Quality of Service (QoS) Guarantees ?			
		Bandwidth	Loss	Order	Timing
Internet	best effort	none	no	no	no
ATM	Constant Bit Rate	Constant rate	yes	yes	yes
ATM	Available Bit Rate	Guaranteed min	no	yes	no
Internet	Intserv Guaranteed (RFC 1633)	yes	yes	yes	yes
Internet	Diffserv (RFC 2475)	possible	possibly	possibly	no



THANK YOU

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